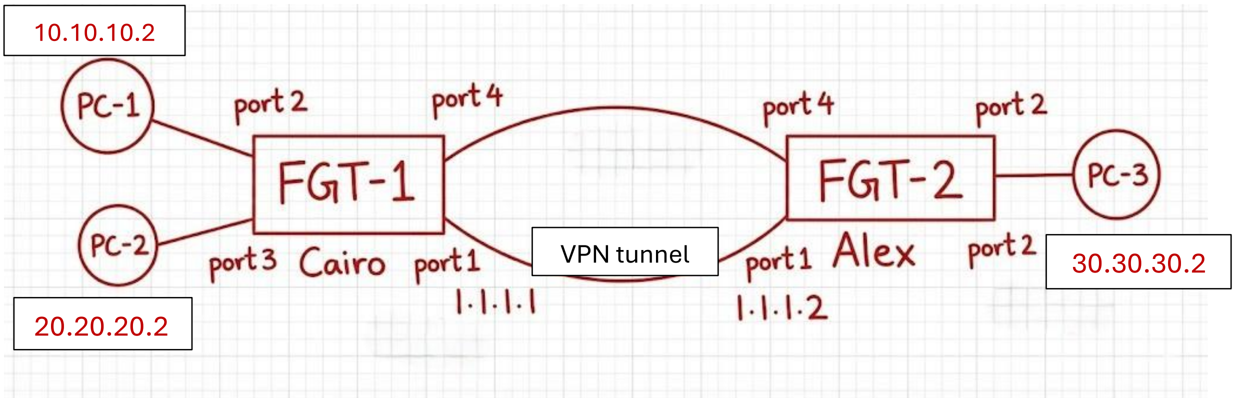
Week 2: IPsec VPN Configuration

Task: Establish an IPsec VPN tunnel between two FortiGate devices.

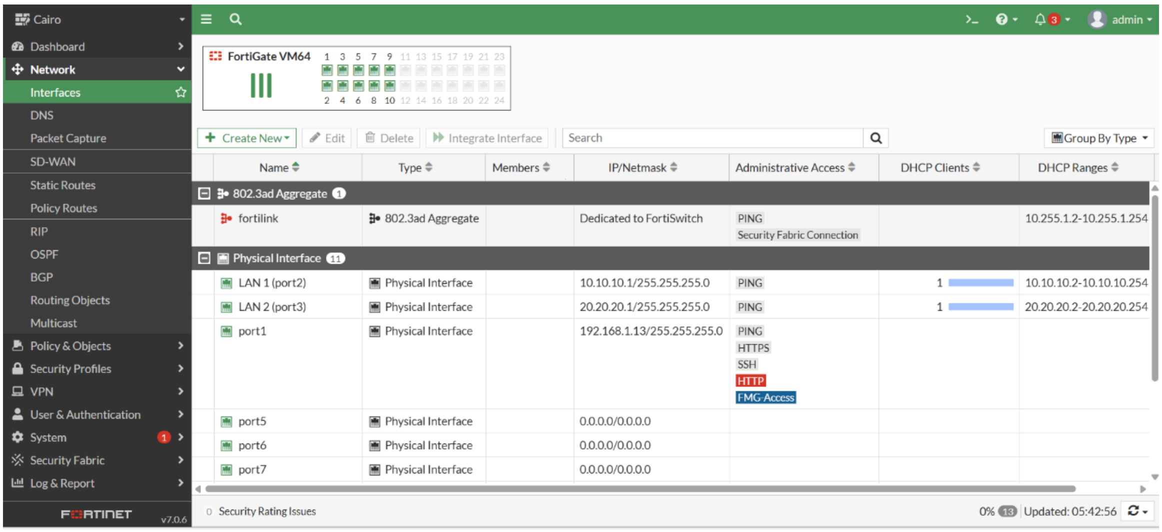
We established this topology:



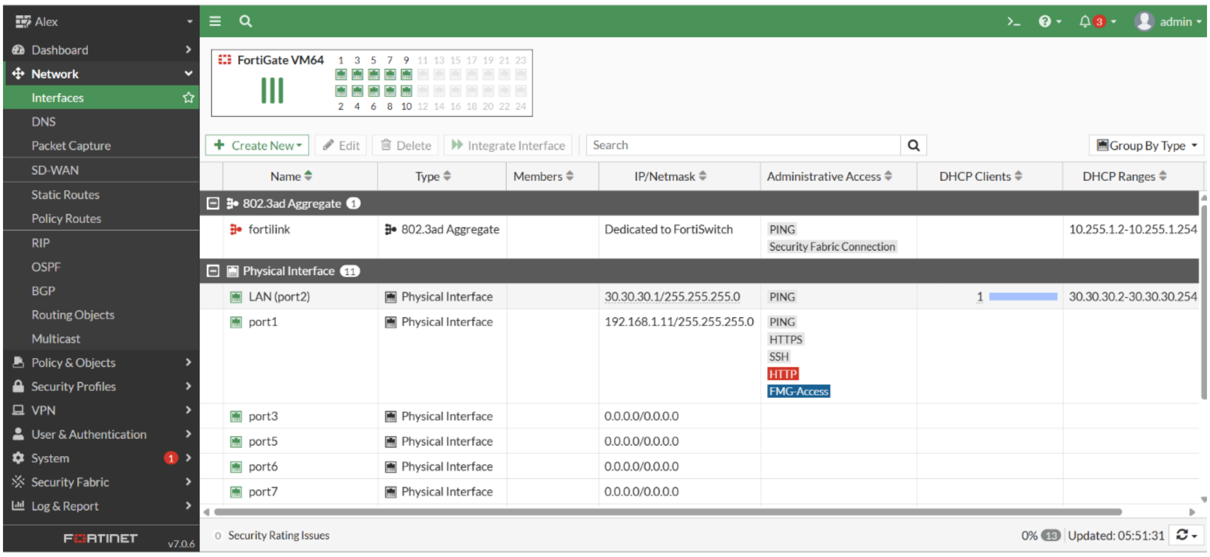
Configuration:

1. Configure the interfaces and make port 1 is the management port

Cairo:

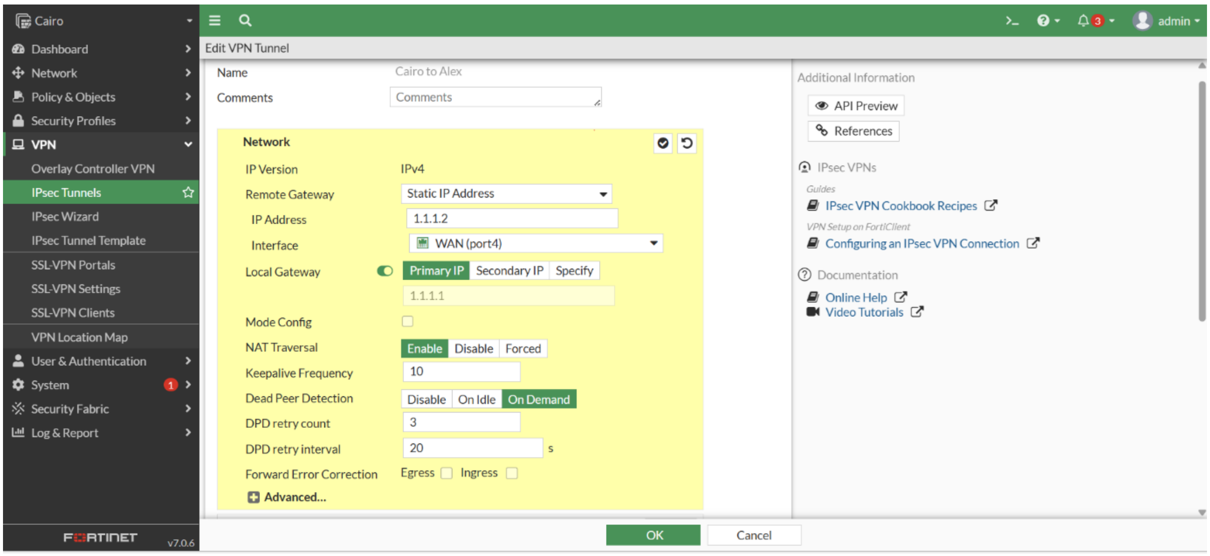


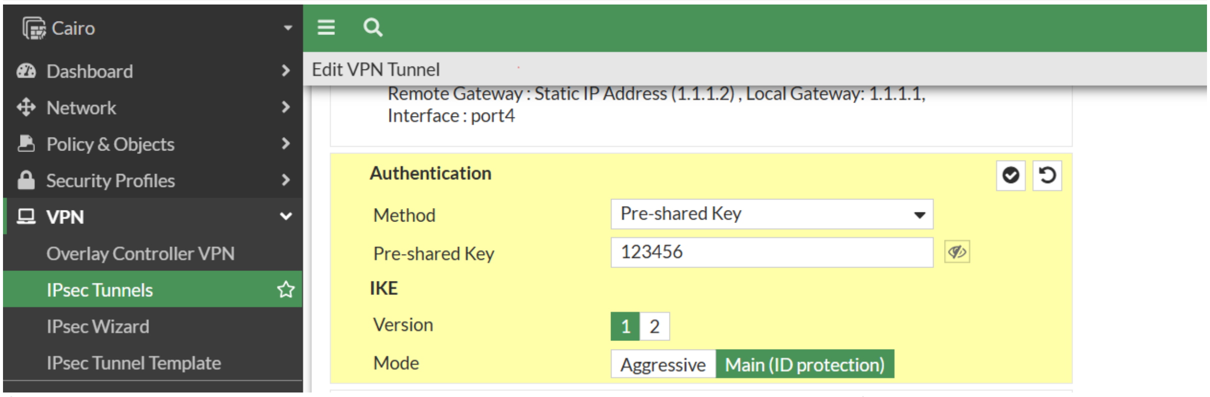
Alex:

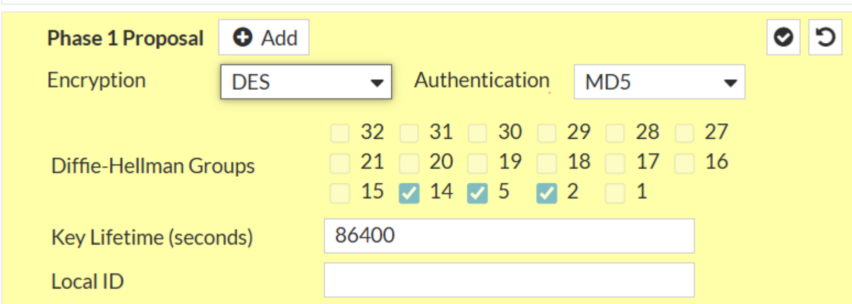


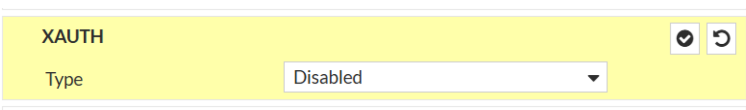
2. Configure the vpn tunnel between the two firewalls and choose traffic selector from the network 10.10.10.0 to 30.30.30.0 in Cairo firewall and reverse in the Alex firewall

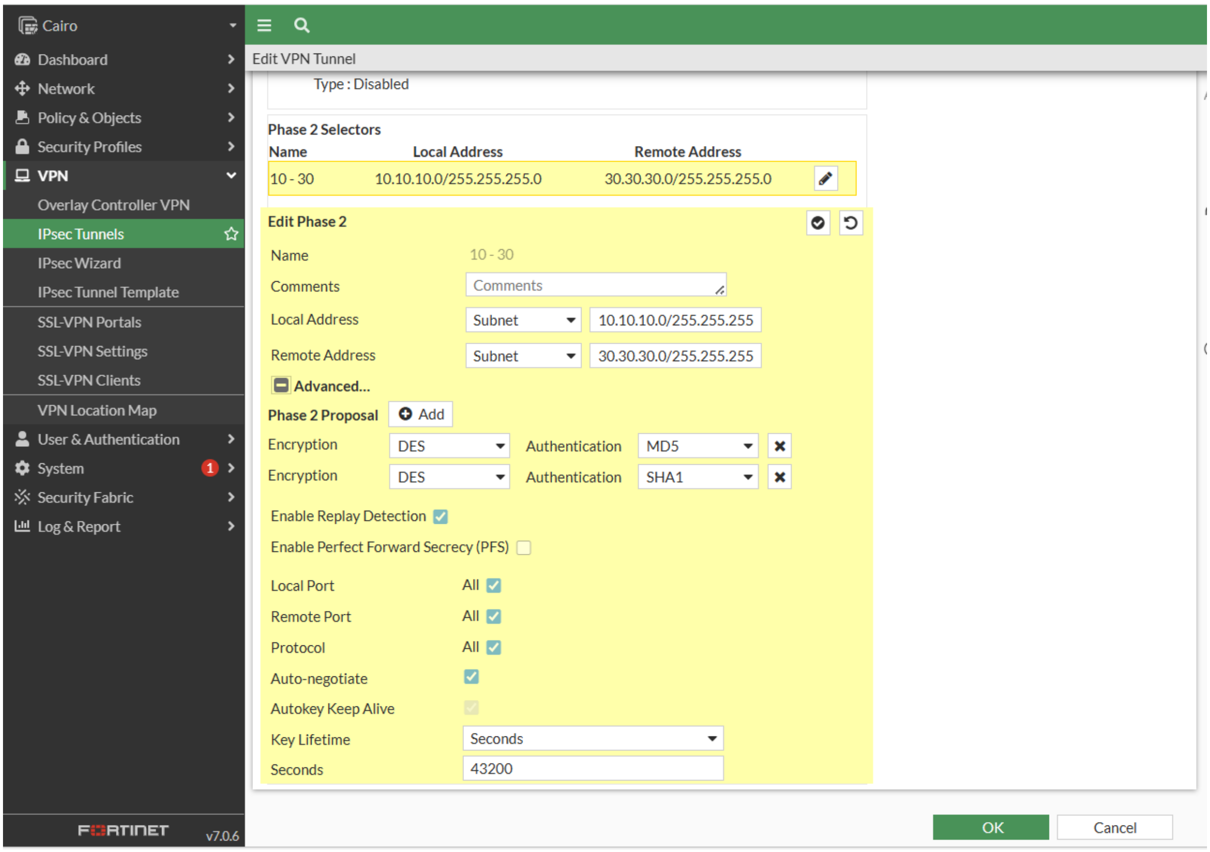
Cairo:

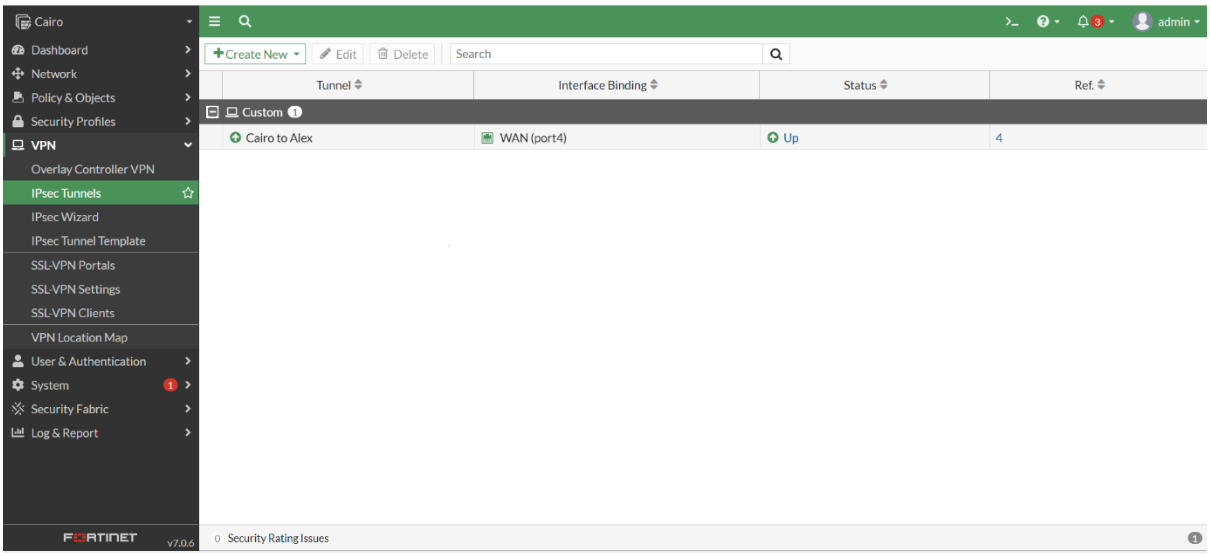




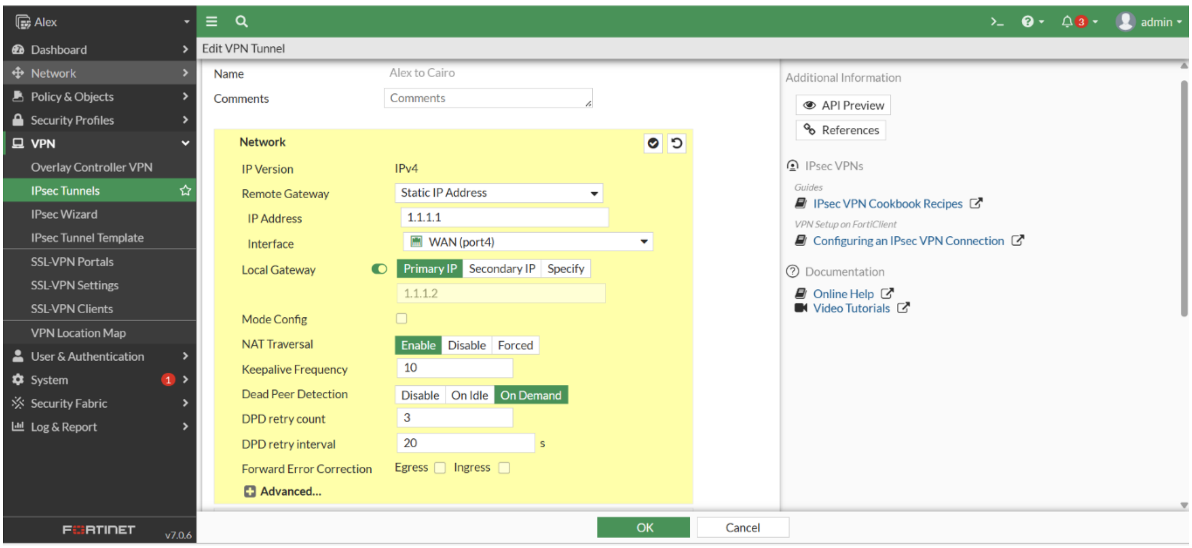


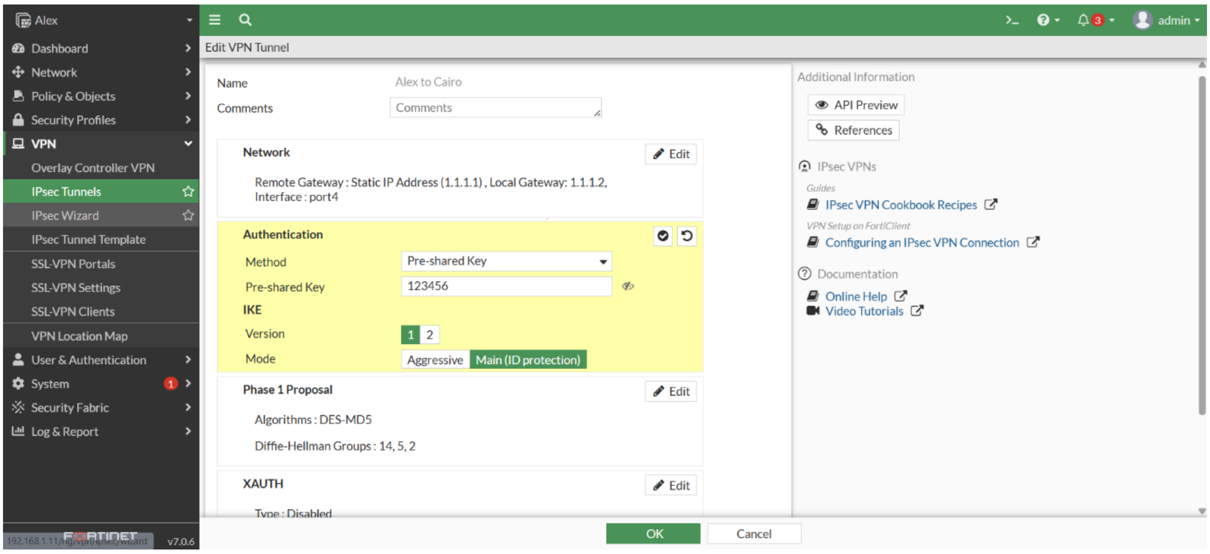


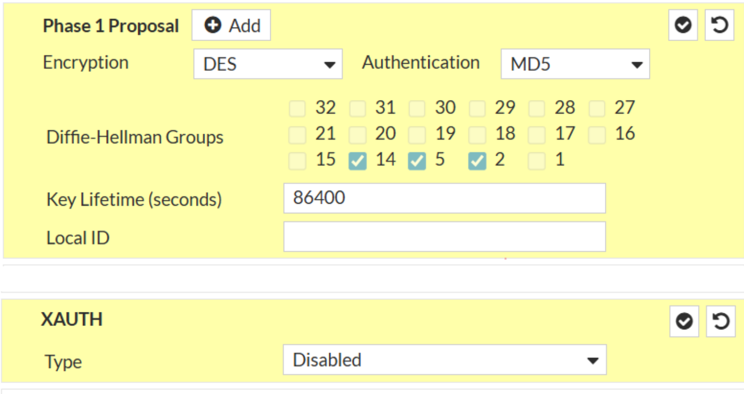

Cairo
Dashboard
Network
Policy & Objects
Security Profiles
VPN
Overlay Controller VPN
IPsec Tunnels
IPsec Wizard
IPsec Tunnel Template
SSL-VPN Portals
SSL-VPN Settings
SSL-VPN Clients
VPN Location Map
User & Authentication
System
Security Fabric
Log & Report
v7.0.6
Edit VPN Tunnel
Type : Disabled
Phase 2 Selectors
Name
Local Address
Remote Address
10 - 30
10.10.10.0/255.255.255.0
30.30.30.0/255.255.255.0
Edit Phase 2
Name
10 - 30
Comments
Local Address
Subnet
10.10.10.0/255.255.255
Remote Address
Subnet
30.30.30.0/255.255.255
Advanced...
Phase 2 Proposal
Add
Encryption
DES
Authentication
MD5
Encryption
DES
Authentication
SHA1
Enable Replay Detection
Enable Perfect Forward Secrecy (PFS)
Local Port
All
Remote Port
All
Protocol
All
Auto-negotiate
Autokey Keep Alive
Key Lifetime
Seconds
Seconds
43200
OK
Cancel

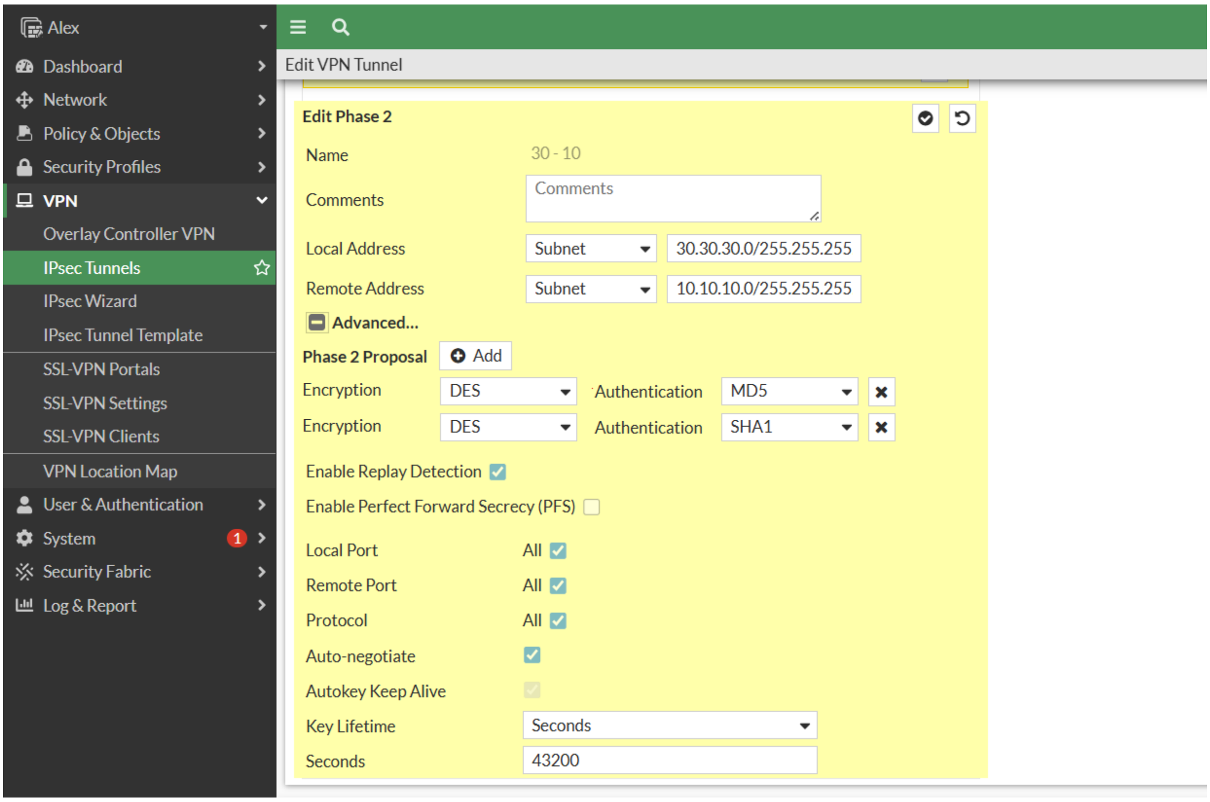

Cairo
admin
Create New
Edit
Delete
Search
Tunnel
Interface Binding
Status
Ref.
Custom
Cairo to Alex
WAN (port4)
Up
4
Security Rating Issues

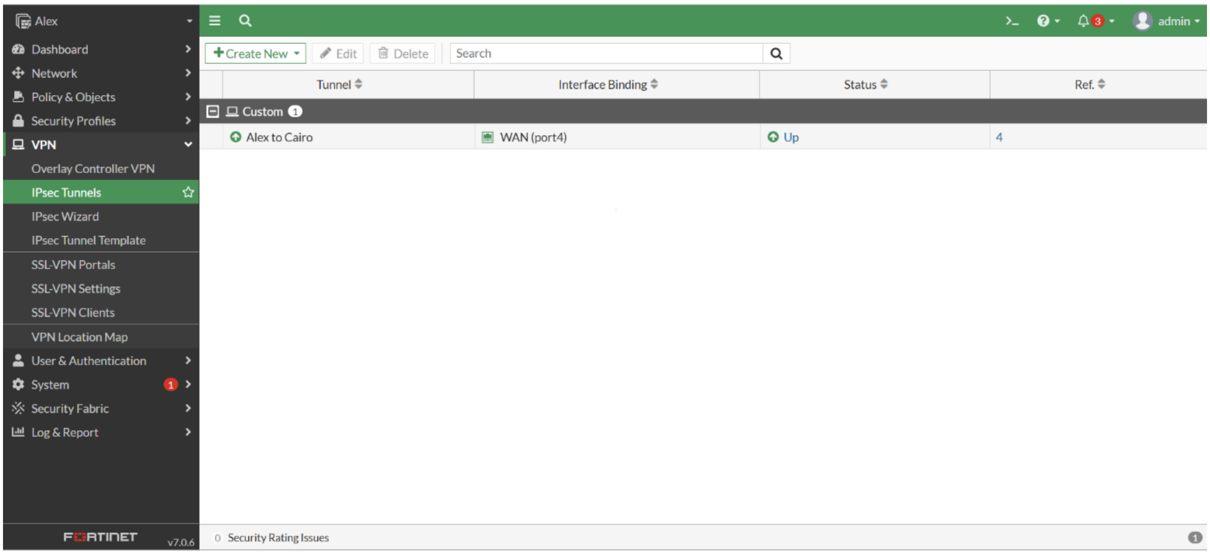
Alex:


Alex
admin
Edit VPN Tunnel
Name
Alex to Cairo
Comments
Network
IP Version
IPv4
Remote Gateway
Static IP Address
IP Address
1.1.1.1
Interface
WAN (port4)
Local Gateway
Primary IP
Secondary IP
Specify
1.1.1.2
Mode Config
NAT Traversal
Enable
Disable
Forced
Keepalive Frequency
10
Dead Peer Detection
Disable
On Idle
On Demand
DPD retry count
3
DPD retry interval
20
Forward Error Correction
Egress
Ingress
Advanced...
Additional Information
API Preview
References
IPsec VPNs
IPsec VPN Cookbook Recipes
Configuring an IPsec VPN Connection
Documentation
Online Help
Video Tutorials
OK
Cancel


Alex
Dashboard
Network
Policy & Objects
Security Profiles
VPN
Overlay Controller VPN
IPsec Tunnels
IPsec Wizard
IPsec Tunnel Template
SSL-VPN Portals
SSL-VPN Settings
SSL-VPN Clients
VPN Location Map
User & Authentication
System
Security Fabric
Log & Report
v7.0.6
admin
Edit VPN Tunnel
Name
Alex to Cairo
Comments
Comments
Network
Edit
Remote Gateway : Static IP Address (1.1.1.1) , Local Gateway: 1.1.1.2,
Interface : port4
Authentication
Method
Pre-shared Key
Pre-shared Key
123456
IKE
Version
1
2
Mode
Aggressive
Main (ID protection)
Phase 1 Proposal
Edit
Algorithms : DES-MD5
Diffie-Hellman Groups : 14, 5, 2
XAUTH
Edit
Type : Disabled
OK
Cancel
Additional Information
API Preview
References
IPsec VPNs
Guides
IPsec VPN Cookbook Recipes
VPN Setup on FortiClient
Configuring an IPsec VPN Connection
Documentation
Online Help
Video Tutorials

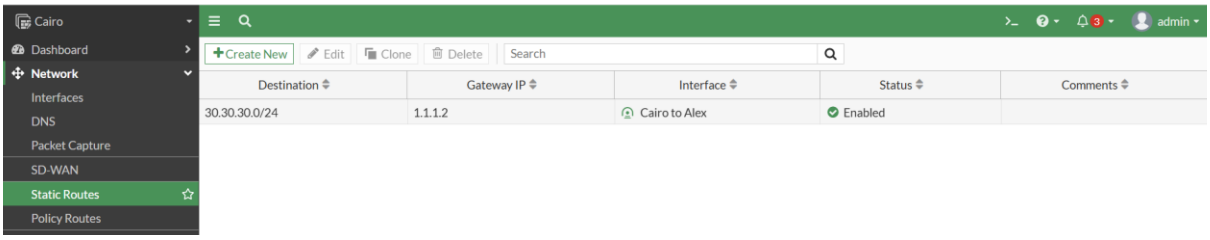

Phase 1 Proposal
Add
Encryption
DES
Authentication
MD5
Diffie-Hellman Groups
32 31 30 29 28 27
21 20 19 18 17 16
15 14 5 2 1
Key Lifetime (seconds)
86400
Local ID
XAUTH
Type
Disabled


Alex
Dashboard
Network
Policy & Objects
Security Profiles
VPN
Overlay Controller VPN
IPsec Tunnels
IPsec Wizard
IPsec Tunnel Template
SSL-VPN Portals
SSL-VPN Settings
SSL-VPN Clients
VPN Location Map
User & Authentication
System
Security Fabric
Log & Report
Edit VPN Tunnel
Edit Phase 2
Name
30 - 10
Comments
Comments
Local Address
Subnet
30.30.30.0/255.255.255
Remote Address
Subnet
10.10.10.0/255.255.255
Advanced...
Phase 2 Proposal
Add
Encryption
DES
Authentication
MD5
Encryption
DES
Authentication
SHA1
Enable Replay Detection
Enable Perfect Forward Secrecy (PFS)
Local Port
All
Remote Port
All
Protocol
All
Auto-negotiate
Autokey Keep Alive
Key Lifetime
Seconds
Seconds
43200

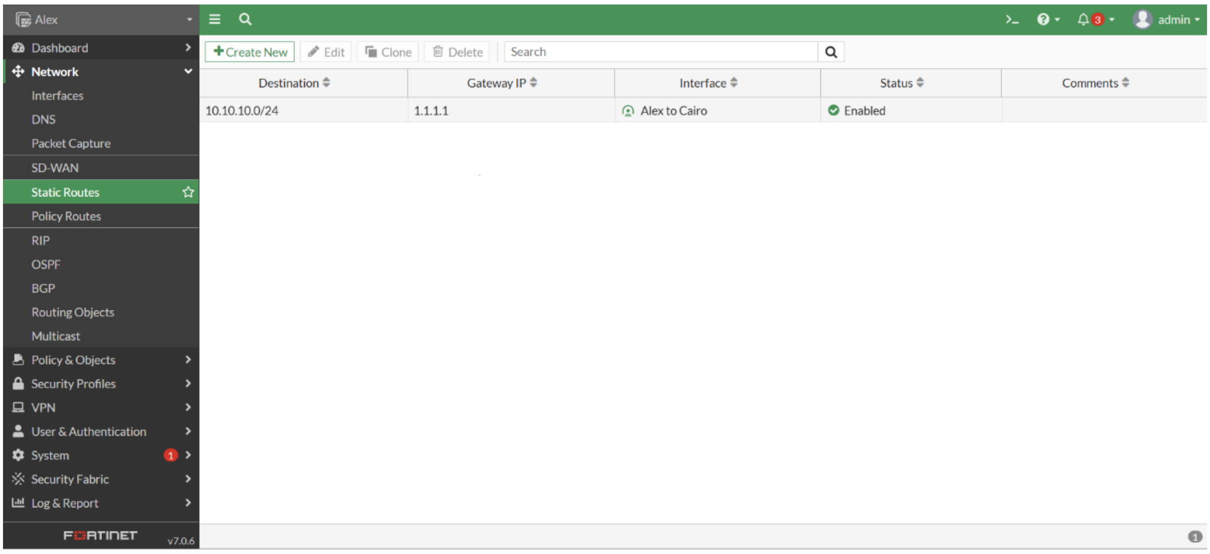


3. Configure the static route

Cairo:

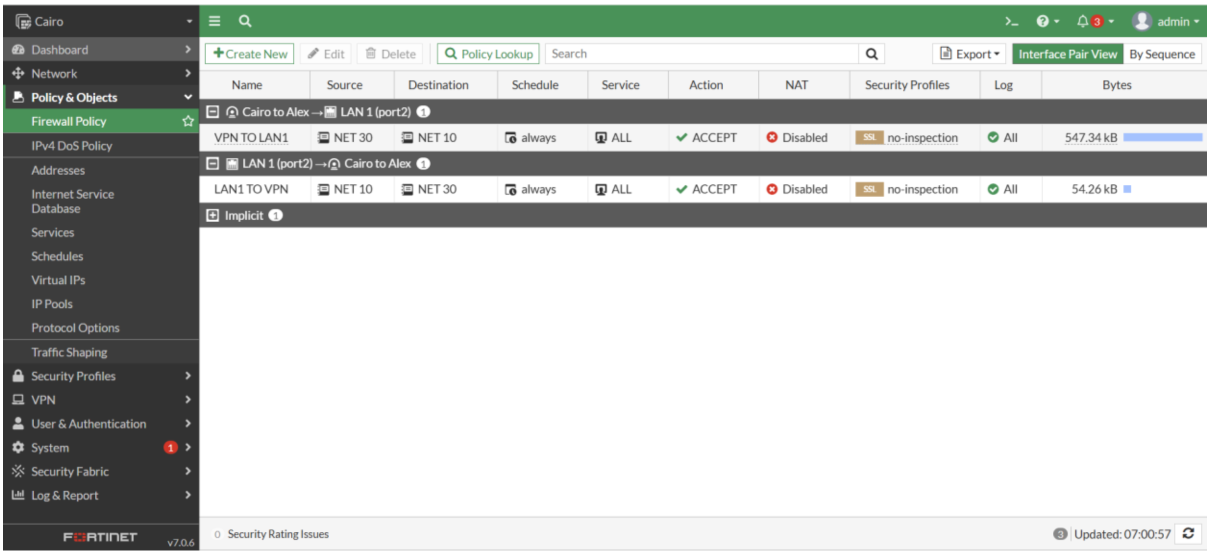


Alex:

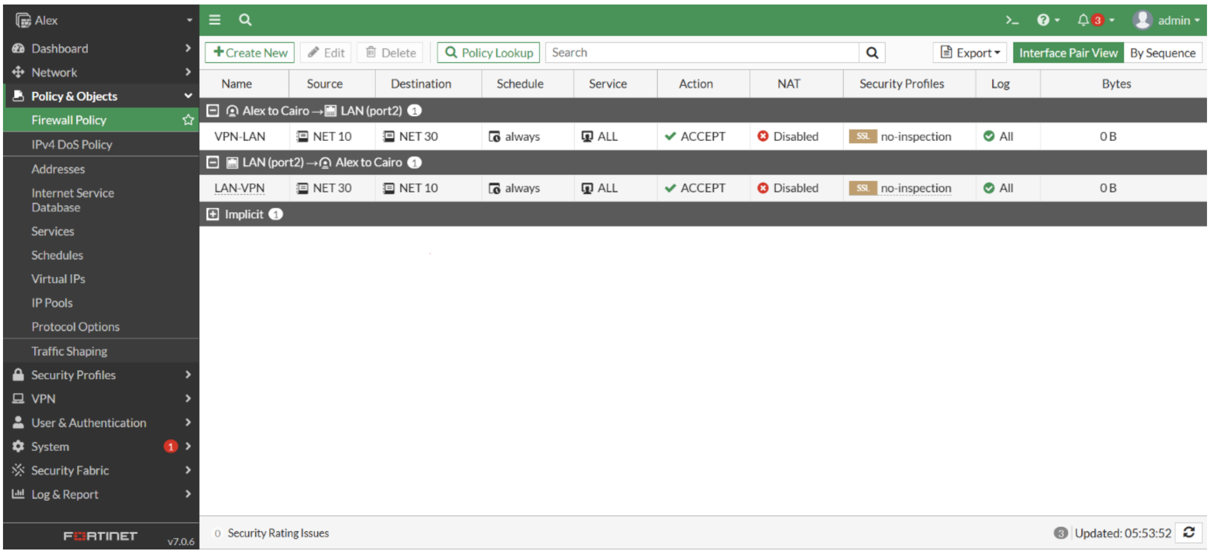


4. Configure the policy between network 10.10.10.0 and 30.30.30.0

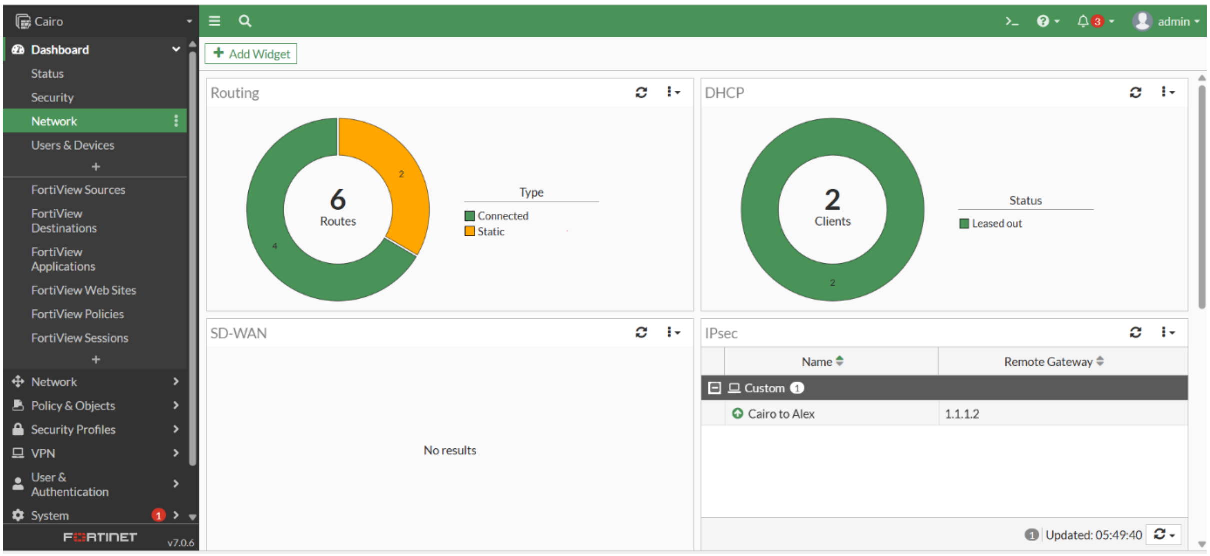
Cairo:

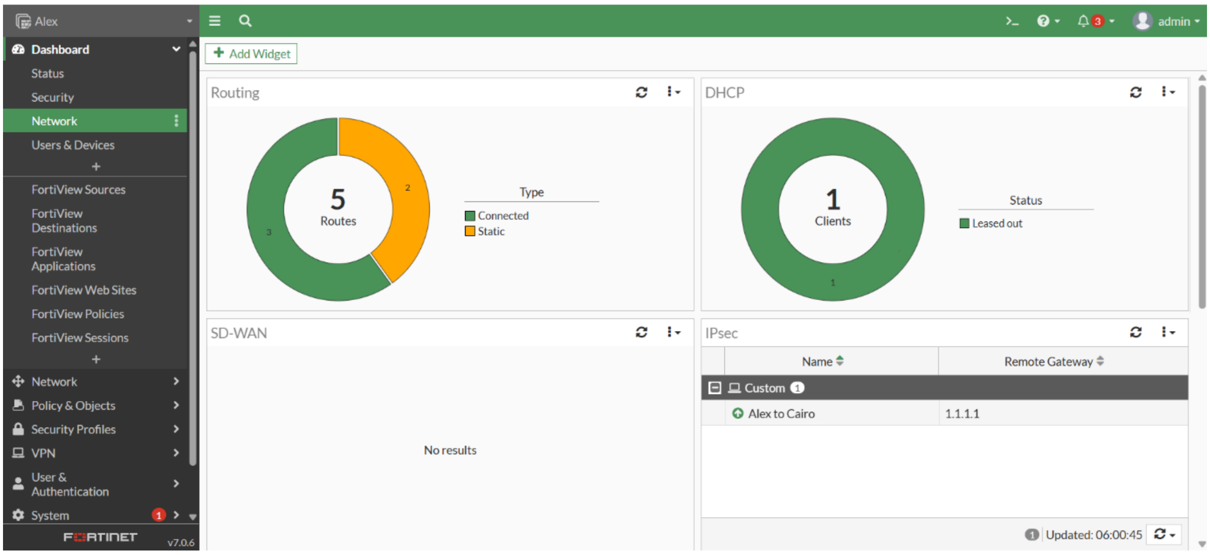


Alex:



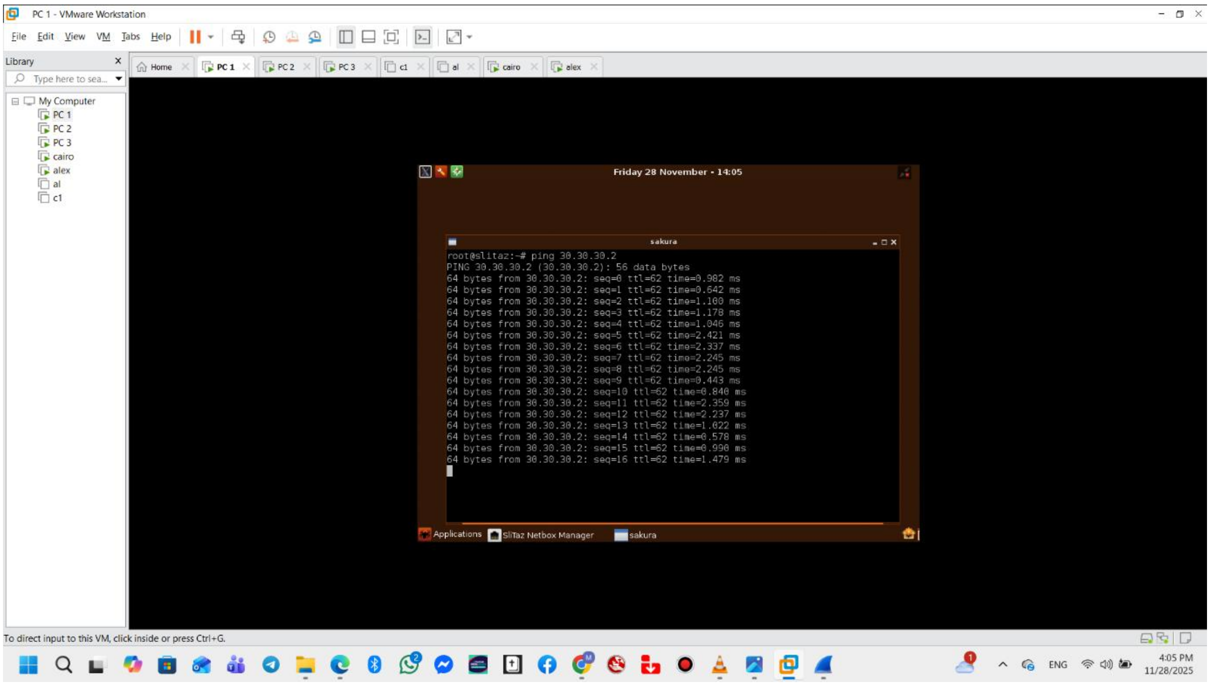
So the Vpn tunnel now is up and running

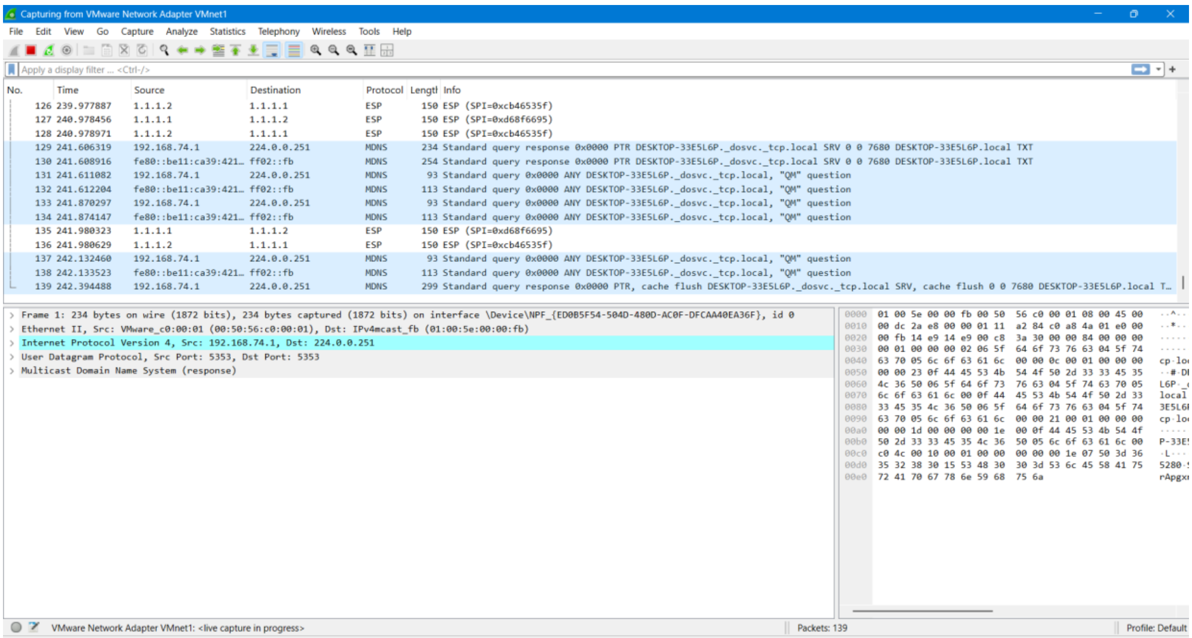




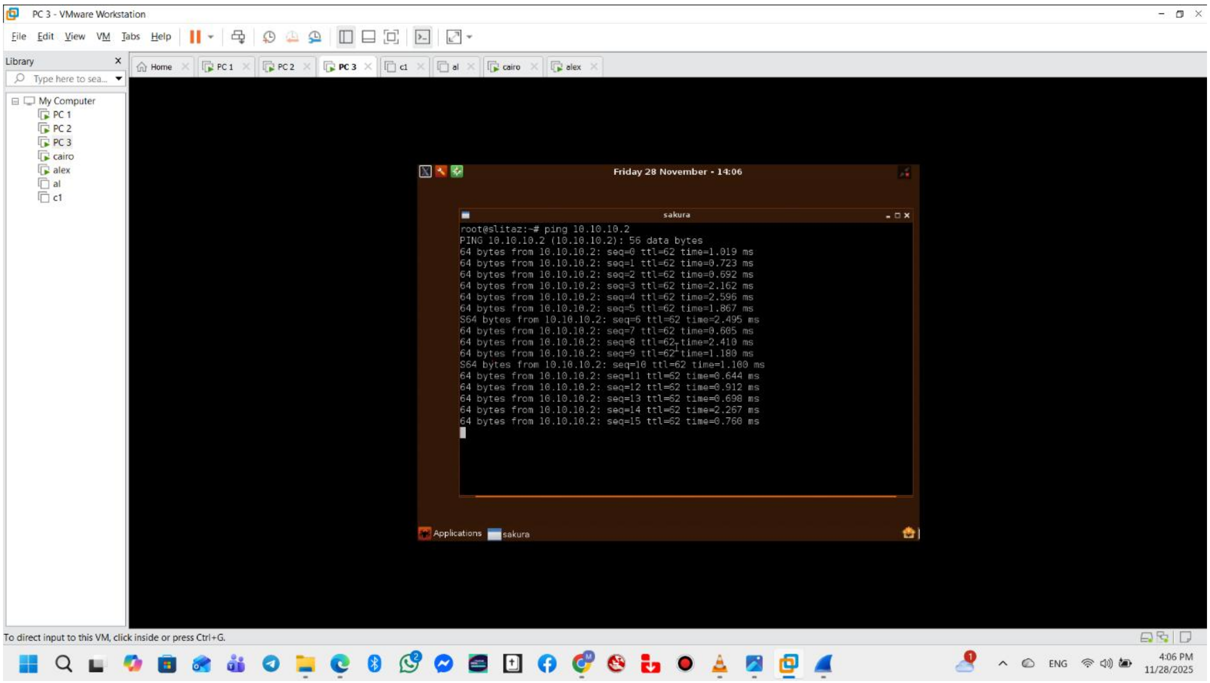
Test Results:

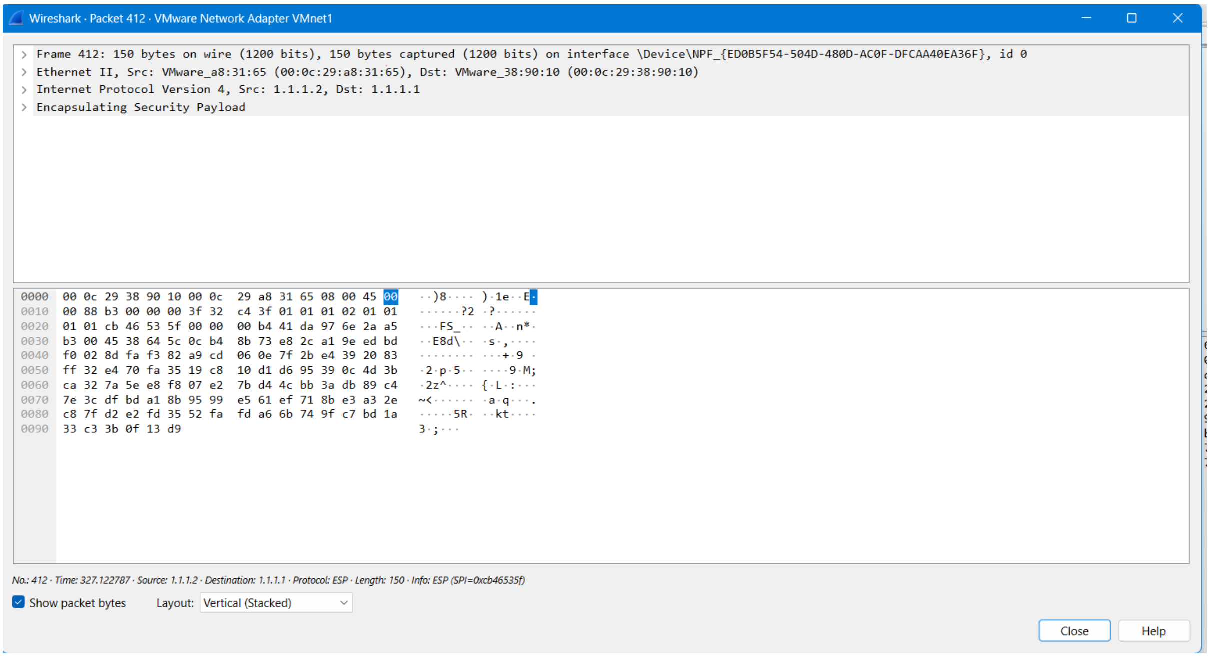
So let us see the result when we ping from PC-1 to PC-3





From PC-3 to PC-1

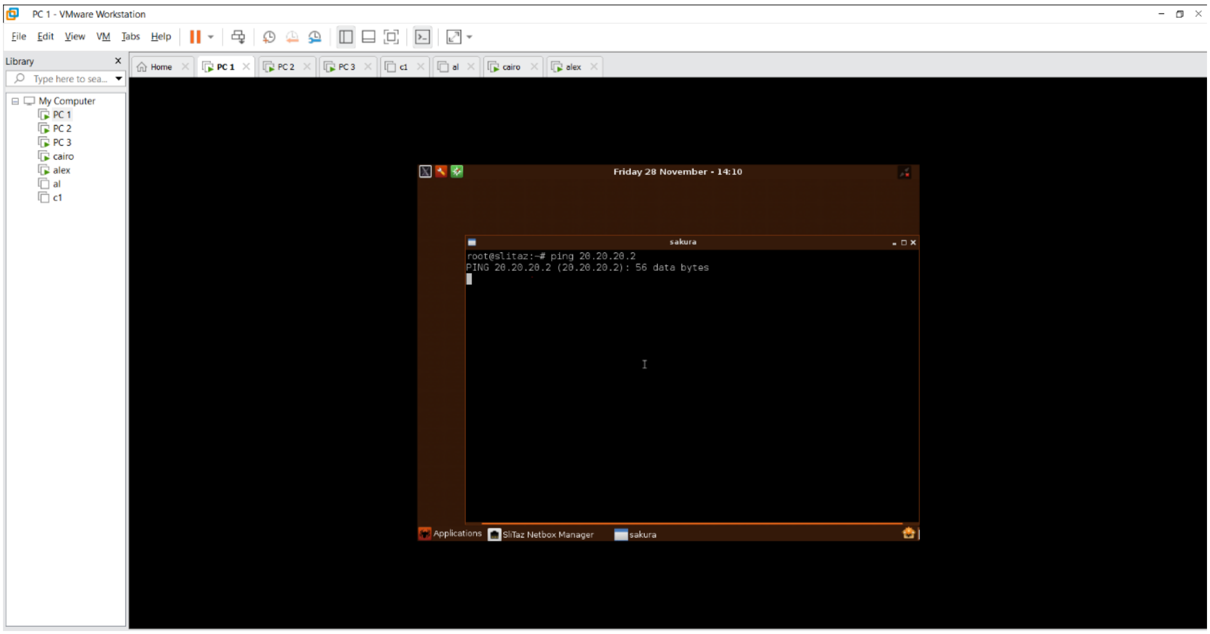




Comment:

The connectivity test results between PC-1 and PC-3 are successful because we established traffic selector from the network 10.10.10.0 to 30.30.30.0

Ping from PC-1 to PC-2



Comment:

The connectivity test results between PC-1 and PC-2 have failed because there is no traffic selector between network 10 and network 20 so the result is as expected